

The nonlocal collisional operators

Their conservative forms and local entropy inequalities

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The Boltzmann equation

The Boltzmann equation for hard spheres describes the statistical behavior of a thermodynamic system out of equilibrium by modeling the time evolution of the distribution function $f(t, x, v)$:

$$\partial_t f + v \cdot \nabla_x f = B[f] = \int_{\mathbb{S}^2 \times \mathbb{R}^3} (f' f'_* - f f_*) \langle v - v_*, n \rangle_+ dn dv_*,$$

where f_* , f' , and f'_* stand for $f(v_*)$, $f(v')$, and $f(v'_*)$ respectively. Moreover the pre-collision velocities (v', v'_*) are given by

$$\begin{cases} v'(v, v_*, n) = v - n \langle n, v - v_* \rangle, \\ v'_*(v, v_*, n) = v_* + n \langle n, v - v_* \rangle. \end{cases}$$

The notation $\langle \cdot \rangle_+$ represents the positive part of the inner product, i.e., $\max\{\langle \cdot \rangle, 0\}$, and $|n| = 1$.

Landau mass current

Theorem (The local conservation laws)

If $f \in L^1(\mathbb{R}^3; (1 + |v|)^4 dv)$, then $B[f] \in L^1(\mathbb{R}^3; (1 + |v|)^2 dv)$, and

$$\int_{\mathbb{R}^3} B[f](v) \begin{pmatrix} 1 \\ v \\ |v|^2 \end{pmatrix} dv = 0.$$

Theorem (C. Villani '99)

In the sense of weak derivatives,

$$B[f] = -\operatorname{div}_v \int_{\langle v-w, n \rangle > 0} \int_0^{\langle v-w, n \rangle} f(t, x, v - rn) f(t, x + an, w - rn) \langle v - w, n \rangle n dr dw dn.$$

The Boltzmann H-theorem

Theorem (H-theorem for hard spheres)

Let $0 < f \in L^1(\mathbb{R}^3; (1 + |v|^2)^{m+1})$ satisfy $|\ln f(v)| \leq C(1 + |v|^2)^m$ for a.e. $v \in \mathbb{R}^3$, then

$$\int_{\mathbb{R}^3} B[f](v) \ln f(v) \, dv \leq 0.$$

Moreover, equality holds if and only if there exists ρ, θ, u such that

$$f(v) = \frac{\rho}{(2\pi\theta)^{3/2}} \exp\left(-\frac{|v - u|^2}{2\theta}\right).$$

As a consequence, we have the entropy inequality

$$\frac{d}{dt} H(t) = \frac{d}{dt} \int f \ln f \, dx \, dv \leq 0.$$

Delocalized collision models

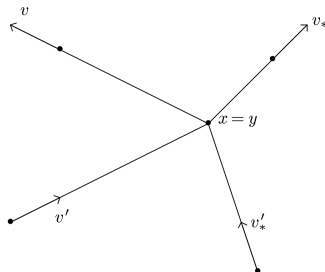
Dilute gas: Boltzmann

Particles

Occupied volume Nr^3

mass points

$O(r)$, negligible

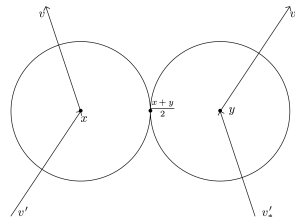


Collision geometry

Dense gas: Enskog

finite-size spheres

$O(1)$, finite volume effect



Delocalized collision integrals

Common structural form

$$\mathcal{C}[f] = \int_{\mathbb{R}^3 \times \mathbb{R}^3} \left(f' f'_* \mathbf{1}_{\langle v-v_*, n_{x,y} \rangle > 0} - f f_* \mathbf{1}_{\langle v-v_*, n_{x,y} \rangle < 0} \right) B[\rho_f, \rho_f](x, y, v - v_*) \, dy \, dv_*.$$

Here $\rho_\varphi := \int \varphi(v) \, dv$, and $n_{x,y} := \frac{x-y}{|x-y|}$. The kernel satisfies symmetry, invariance under the collision map, finite spatial range, and suitable integrability in (x, y) .

Examples covered

Enskog–Boltzmann: $B[\rho, \varpi](x, y, v - v_*) = |\langle v - v_*, n_{x,y} \rangle| \delta_{2r}(|x - y|),$

Soft spheres: $B[\rho, \varpi](x, y, v - v_*) = \theta_{\epsilon, r}(|x - y|) |\langle v - v_*, n_{x,y} \rangle|,$

Povzner-type: $B[\rho, \varpi](x, y, v - v_*) = b(|x - y|) |\langle v - v_*, n_{x,y} \rangle|,$

Revised Enskog: $B[\rho, \varpi](x, y, v - v_*) = Y[\rho(x), \varpi(y)] |\langle v - v_*, n_{x,y} \rangle| \delta_{2r}(|x - y|).$

Motivation

For delocalized collision models, we face a significant challenge: while mass is conserved locally, the momentum and energy are no longer conserved locally. Instead, we can only write the global conservation law:

$$\int_{\mathbb{R}^3 \times \mathbb{R}^3} \mathcal{C}[f](x, v) \varphi_i(v) \, dv \, dx = 0,$$

where $\varphi_i(v)$ are $1, v_1, v_2, v_3$, and $|v|^2/2$ for $i = 0, 1, 2, 3, 4$ respectively.

This leads to a natural question: is there an analogous structure for collision integrals with spatial delocalization?

Theorem (F. Charles, C., F. Golse 2026, C. 2026+)

For each rapidly decaying $f \in C(\mathbb{R}^3 \times \mathbb{R}^3)$, there exist vector fields $\mathcal{I}_k$, and $\mathcal{J}_k$ for $k = 0, 1, 2, 3, 4$

$$\mathcal{C}[f](x, v)\varphi_k(v) = \operatorname{div}_x \mathcal{J}_k + \operatorname{div}_v \mathcal{I}_k,$$

where $\varphi_k(v)$ are 1, v_1 , v_2 , v_3 , and $|v|^2/2$ for $k = 0, 1, 2, 3, 4$ respectively.

A direct application is that we can derive the macroscopic equations without scaling or fluctuation around equilibrium.

Local entropy inequality

Theorem (F. Charles, C., F. Golse 2026)

Assume that $f(t, x, v)$ is a classical solution of

$$(\partial_t + v \cdot \nabla_x) f(t, x, v) = \mathcal{C}[f](t, x, v),$$

with $(x, v) \mapsto f(t, x, v)$ rapidly decaying at infinity, while $(x, v) \mapsto \ln f(t, x, v)$ has at most polynomial growth at infinity. Then, in the sense of distributions,

$$\begin{aligned} \mathcal{C}[f](t, x, v) \ln f(t, x, v) &\leq -\nabla_x \cdot \mathcal{K}[f](t, x, v) - \nabla_v \cdot \mathcal{L}[f](t, x, v) \\ &\quad - \frac{1}{2} (\partial_t + v \cdot \nabla_x) \left(f(t, x, v) (\rho_f(t, \cdot) * \beta(|\cdot|))(x) \right). \end{aligned}$$

Here $*$ denotes convolution in the variable x :

$$\rho_f(t, \cdot) * \beta(|\cdot|)(x) := \int_{\mathbb{R}^3} \rho_f(t, x - y) \beta(|y|) dy.$$

H-theorem

By integrating the local entropy inequality in both x and v , we can obtain the H-theorem for the nonlocal collision equation

Corollary (F. Charles, C., F. Golse 2026)

Assume that $f(t, x, v)$ is a classical solution of

$$(\partial_t + v \cdot \nabla_x) f(t, x, v) = \mathcal{C}[f](t, x, v),$$

with the same decay assumptions as in the local entropy inequality. Then

$$\frac{d}{dt} \mathcal{H}[f](t) = -\Lambda[f](t) \leq 0, \quad \mathcal{H}[f] := H_B[f] + H_\beta[\rho_f].$$

The two parts of the entropy are

$$H_B[f](t) := \int_{\mathbb{R}^3 \times \mathbb{R}^3} f \ln f \, dx \, dv, \quad H_\beta[\rho_f](t) := \frac{1}{2} \int_{\mathbb{R}^3 \times \mathbb{R}^3} \rho_f(t, x) \rho_f(t, y) \beta(|x - y|) \, dx \, dy.$$

Thanks for your attention